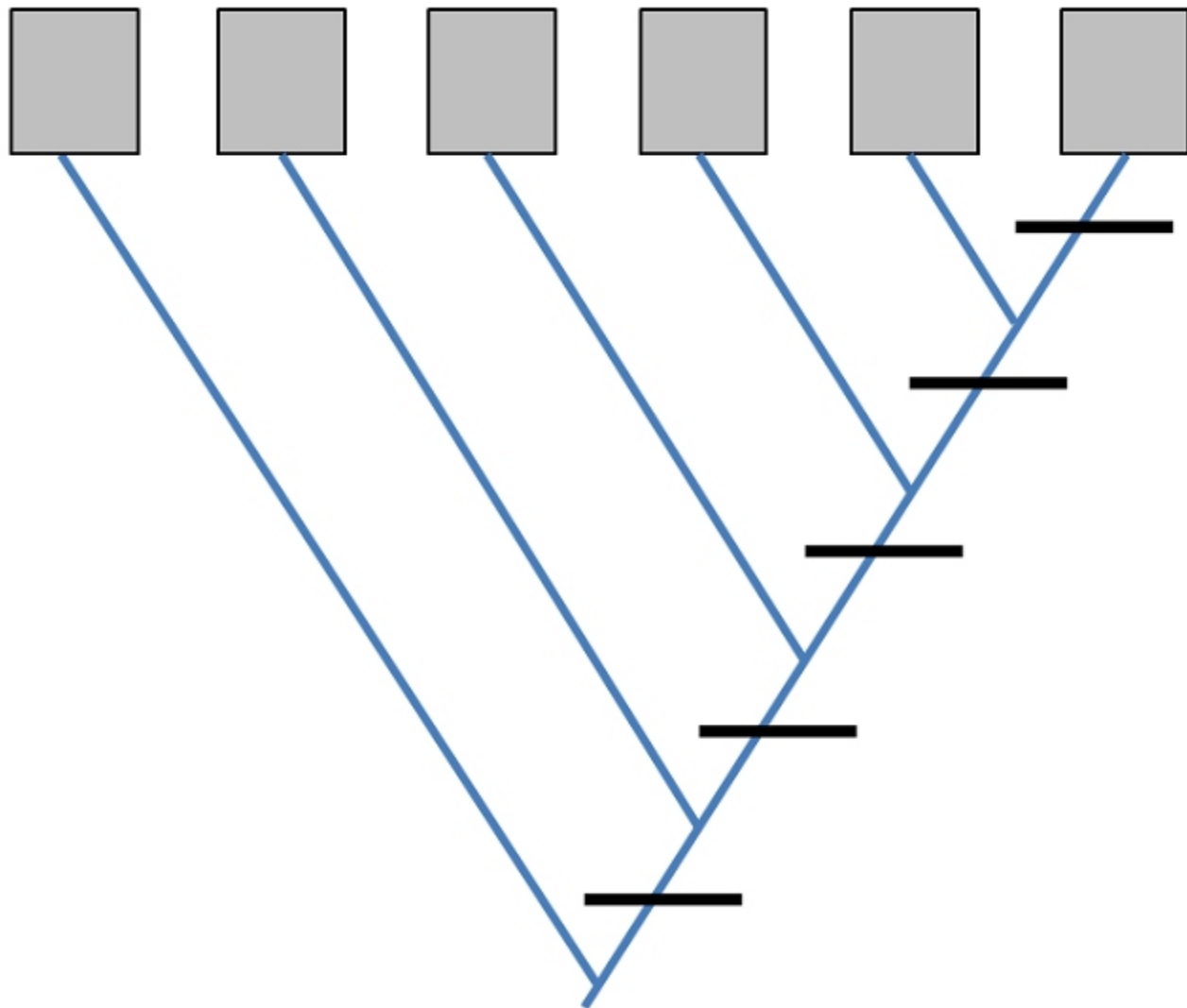




	Thick Hooves (1)	Big Teeth (2)	Soft Fur (3)	Squeaky Voice (4)	Red Nose (5)
Dasher (D)	Y	N	Y	N	N
Prancer (P)	N	Y	N	Y	N
Vixen (V)	N	Y	N	N	N
Comet (C)	Y	N	N	N	N
Rudolph (R)	Y	N	Y	N	Y
Blitzen (B)	Y	Y	N	N	N

Based on the information in the table below, construct a cladogram (or family tree) that depicts the relatedness of Christmas reindeer based on ancestral or derived characters. **Place each of the reindeer "D, P, V, C and B" in the gray boxes on the terminal branches of the tree. Also, place the character state using the character numbers (1-5) on the black dashes which intersect the tree.** Be sure to indicate whether these characters **APPEARED** or **DISAPPEARED** at that intersection on the tree using a + or – symbol (i.e. 1+ or 2-).

*** For cladistic purposes, Rudolph will be considered the OUTGROUP since he is the OLDEST